

PROJECT 4 OF 8 — PART 2: FRAMEWORKS + EXECUTIVE SUMMARY

COLONIAL PIPELINE RANSOMWARE (2021)

Ransomware · No MFA · East Coast Fuel Crisis · \$4.4M Ransom Paid

 **THIS PROJECT:** Apply the NIST CSF and pentesting methodology. Write and present your professional Executive Summary. This project ends with your Executive Summary submission (100 points).

PROJECT 4 · DAY 6

MONDAY — NIST CSF — DEFENSIVE FAILURE ANALYSIS

1-2 HRS

 **OBJECTIVE:** Students apply all 5 NIST Cybersecurity Framework functions to identify exactly what the organization did wrong and what they should have done.

PROJECT 4 · DAY 7

TUESDAY — 5 PHASES OF PENTESTING — ATTACKER METHODOLOGY

1-2 HRS

 **OBJECTIVE:** Students map the attacker's actions to the 5 pentesting phases. Think like an attacker to become a better defender.

PROJECT 4 · DAY 8

WEDNESDAY — RECOMMENDATIONS WORKSHOP — "FROM HERE ON OUT"

1-2 HRS

 **OBJECTIVE:** Students develop and workshop 5 specific, actionable recommendations. Peer collaboration to pressure-test each recommendation.

PROJECT 4 · DAY 9

THURSDAY — EXECUTIVE SUMMARY PRESENTATIONS + SUBMISSION (100 PTS DUE)

1-2 HRS

 **OBJECTIVE:** Students complete peer review, deliver professional executive summary presentations, submit, and debrief the full incident.

"FROM HERE ON OUT" — RECOMMENDATIONS REFERENCE

 **NOTE:** These are the controls that, if in place, would have prevented or significantly reduced this breach.

- **MANDATE:** MFA on 100% of remote access systems — VPN, RDP, cloud portals, everything, no exceptions
- **MANDATE:** Dark web monitoring — continuously check if employee credentials appear in breach databases
- **MANDATE:** OT/IT complete network separation — pipeline controls must never share a network with business IT
- **IMPLEMENT:** Offline, air-gapped backups — ransomware cannot encrypt what it cannot reach
- **IMPLEMENT:** Incident response plan specific to ransomware — tested with tabletop exercises annually
- **CONSIDER:** Cyber insurance review — ensure policy covers ransomware payments and business interruption